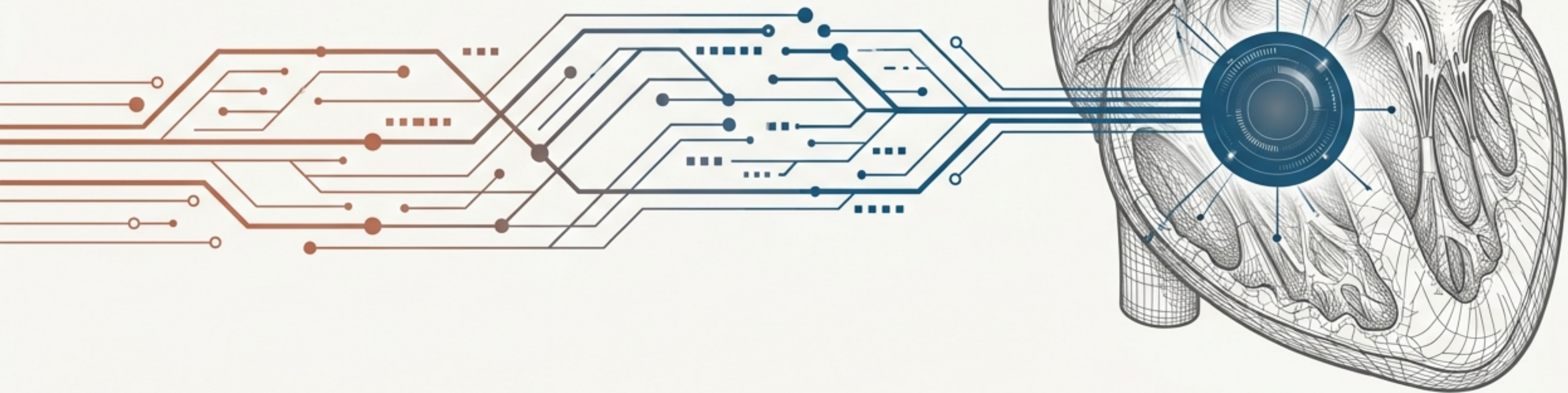


Augmenting Complex Cardiology Care

A Real-World Randomized Controlled Trial
of Large Language Models in Clinical Practice



The Subspecialist Scarcity Crisis

Timely, accurate management determines outcomes in complex cardiology. Yet, access to subspecialist expertise is critically constrained.



18 Million

Projected global deficit of healthcare providers by 2030.

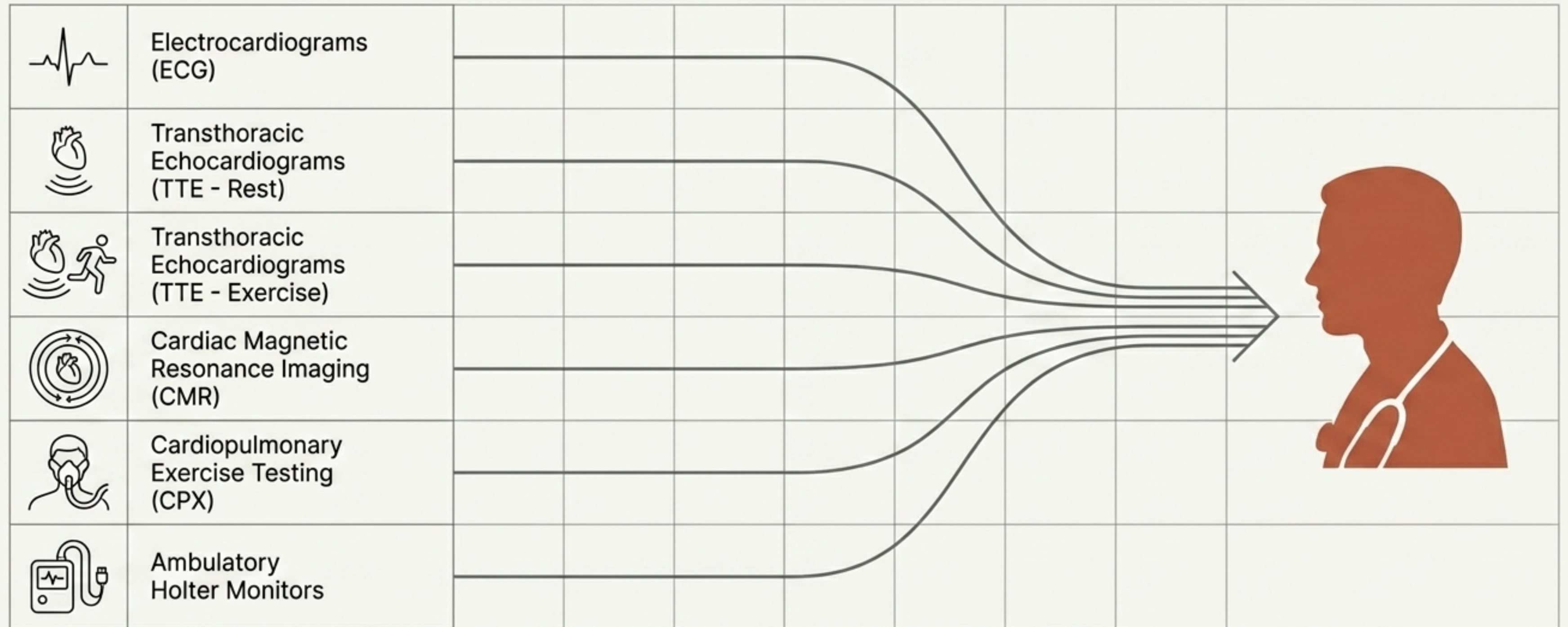
60%

The proportion of hypertrophic cardiomyopathy (HCM) patients currently undiagnosed in the USA.

The Clinical Reality: More than half of US states lack an HCM subspecialist center, leading to missed diagnostic windows for a condition that is a leading cause of preventable sudden cardiac death in young adults.

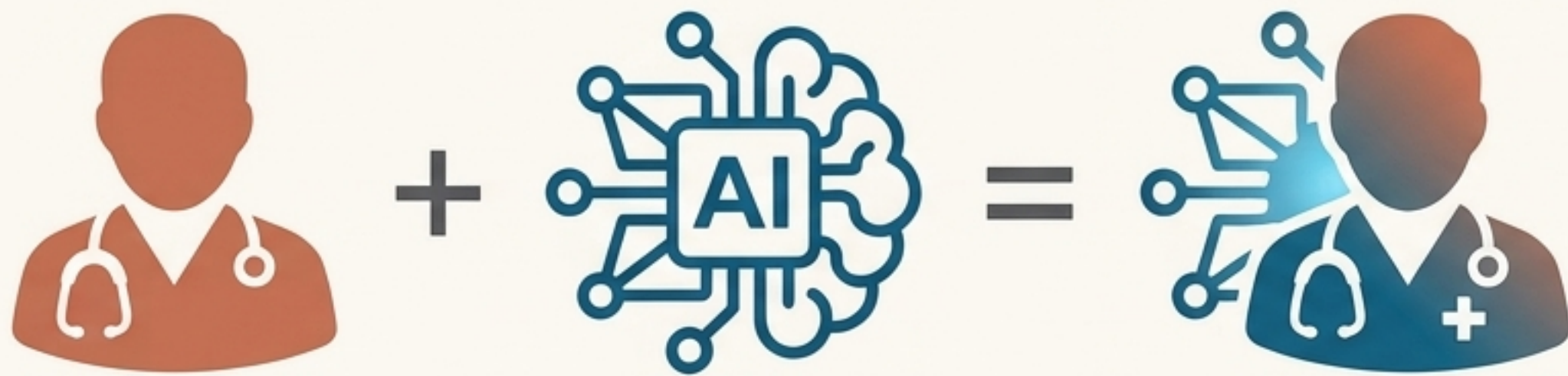
The Multimodal Diagnostic Burden

General cardiologists are tasked with synthesizing disparate, complex data streams to identify rare genetic cardiomyopathies before referring to a specialist.



Enter AMIE: The Augmented Clinician

Articulate Medical Intelligence Explorer (AMIE) is an experimental large language model designed to augment clinical decision-making. Instead of replacing the physician, AMIE is deployed as a collaborative reasoning engine—equipped with multistep inference, web search, and self-critique—to help general cardiologists synthesize multimodal reports and elevate their triage, diagnosis, and management of rare cardiac diseases.



 **AMIE Consultation: Patient ID 703-A (21y/o M)** ...

Age: 21, Sex: M.
Tests: ECG, TTE, Holter, Genetic Panel.
Chief Complaint: Palpitations, family history of SCD.

Physician
AMIE, based on the attached ECG showing deep T-wave inversions in V1-V3 and the TTE report indicating asymmetric septal hypertrophy (22mm) with RV involvement, what is the differential diagnosis?

AMIE
Analyzing ECG and TTE data. The constellation of findings strongly suggests **Arrhythmogenic Right Ventricular Cardiomyopathy (ARVC)** or a specific form of **Hypertrophic Cardiomyopathy (HCM)** with RV involvement. My top differential includes ARVC, Desmoplakin (DSP) cardiomyopathy, or classic HCM with RV apical involvement. Given the family history and RV findings, ARVC/DSP should be considered.

Physician
What specific next steps do you recommend to differentiate and manage, considering the genetic panel is pending?

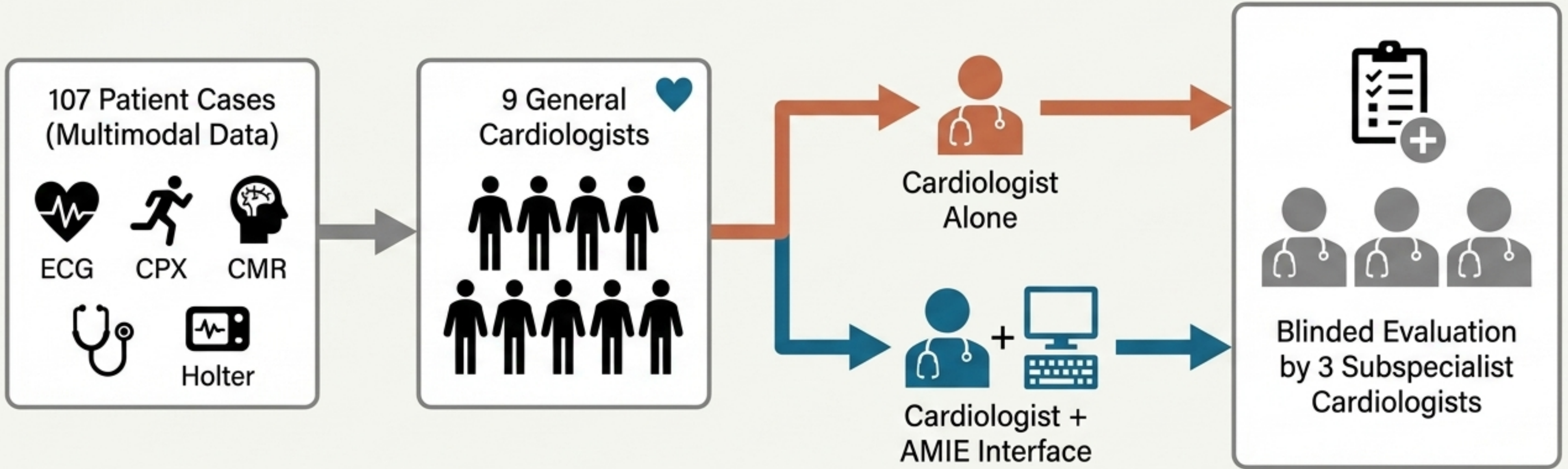
AMIE
To further differentiate, I recommend:

1. **Cardiac Magnetic Resonance (CMR) with LGE:** To visualize RV wall abnormalities, fibrosis, and RV end-diastolic volume, critical for ARVC criteria.
2. **Exercise Stress Test:** To assess for exercise-induced arrhythmias.
3. **Cascade Family Screening:** Initiate genetic counseling and screening for first-degree relatives due to the strong family history.
4. **Refer to EP/HCM Subspecialist:** For specialized evaluation and potential ICD consideration, especially given the NSVT history from Holter.

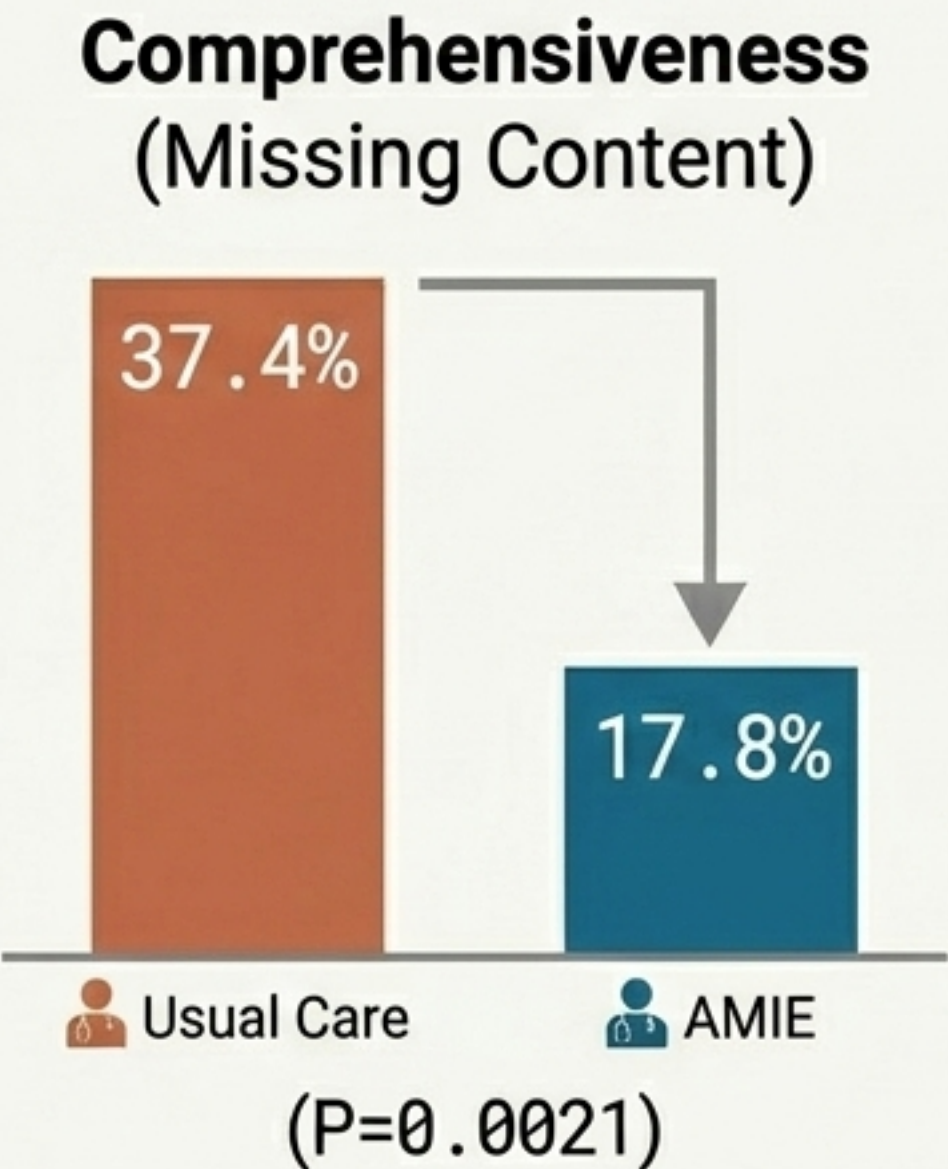
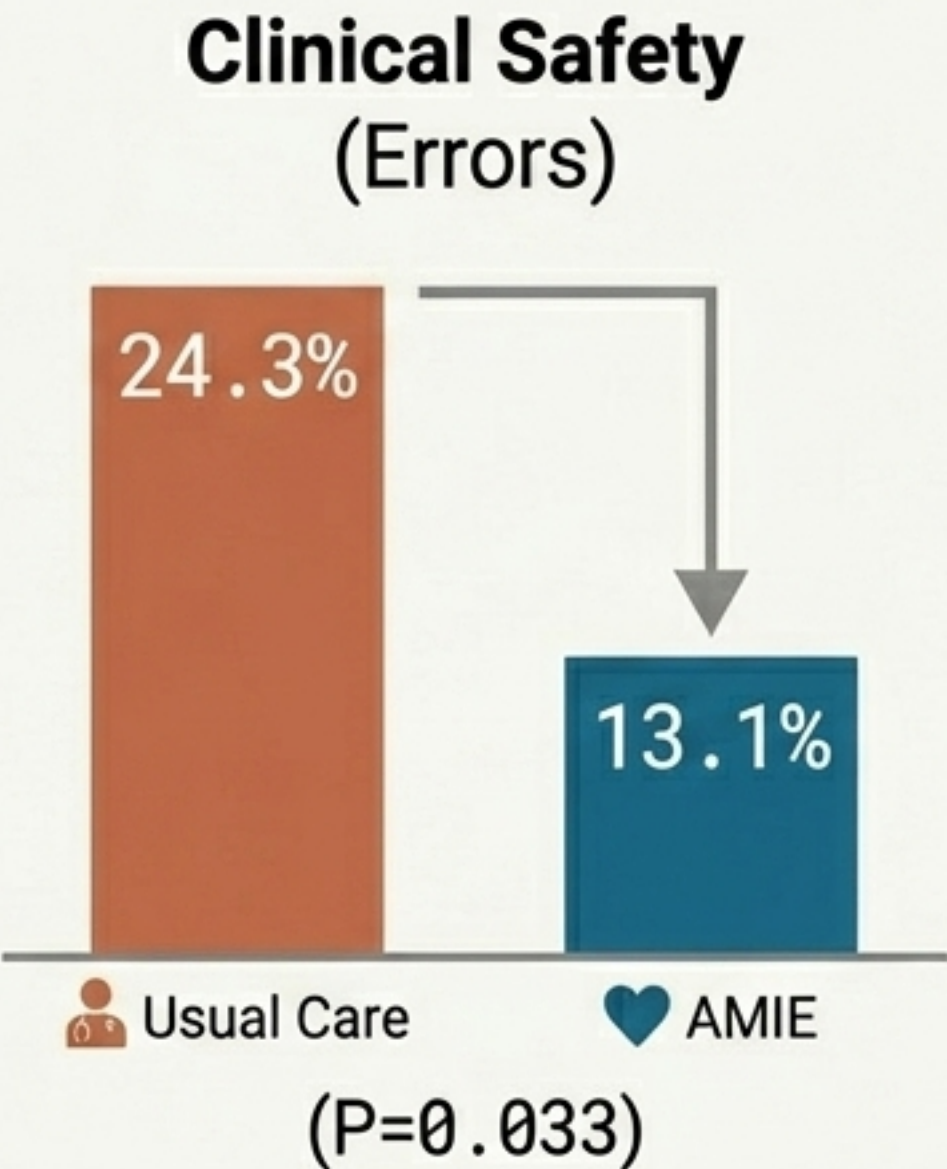
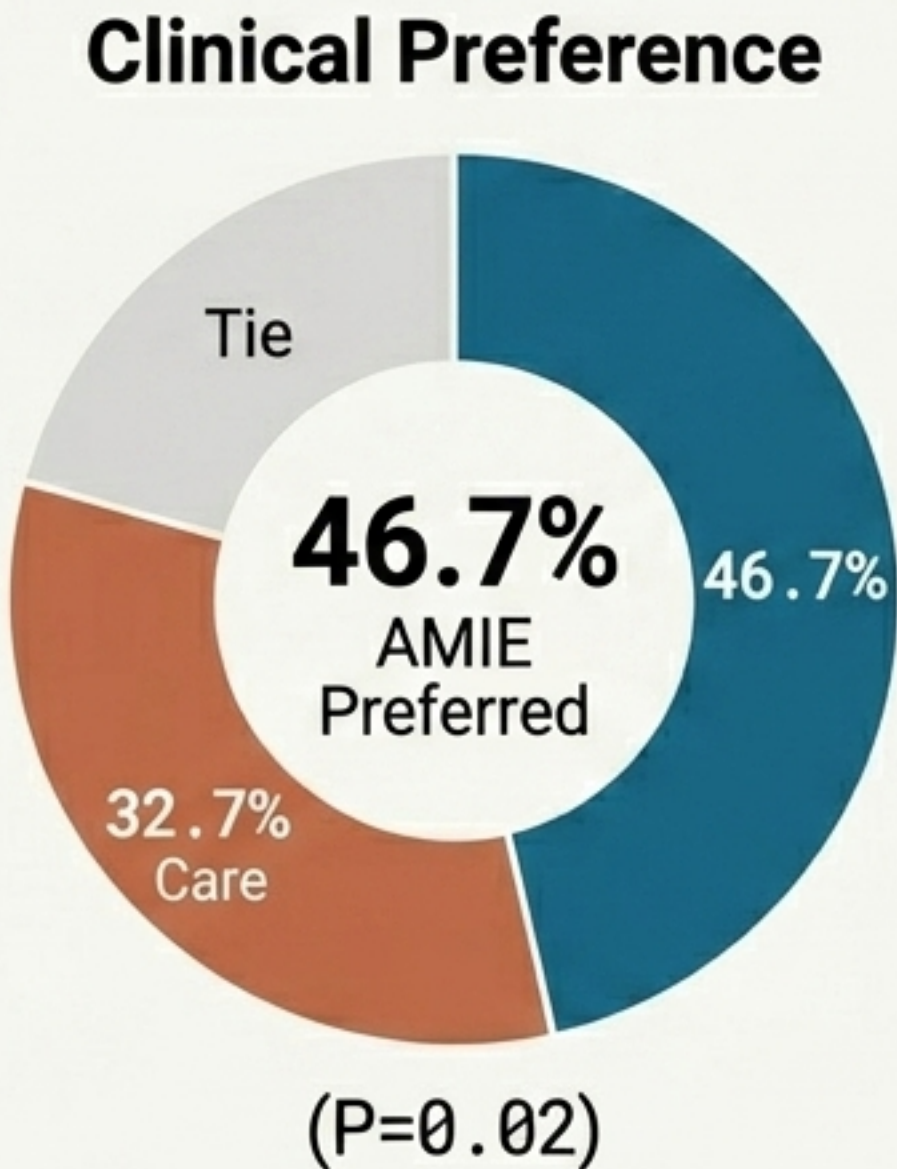
Real-World Randomized Controlled Trial Design

Moving beyond synthetic vignettes, this trial evaluated human performance against human-plus-AI performance using retrospective data from a specialized inherited cardiovascular center.



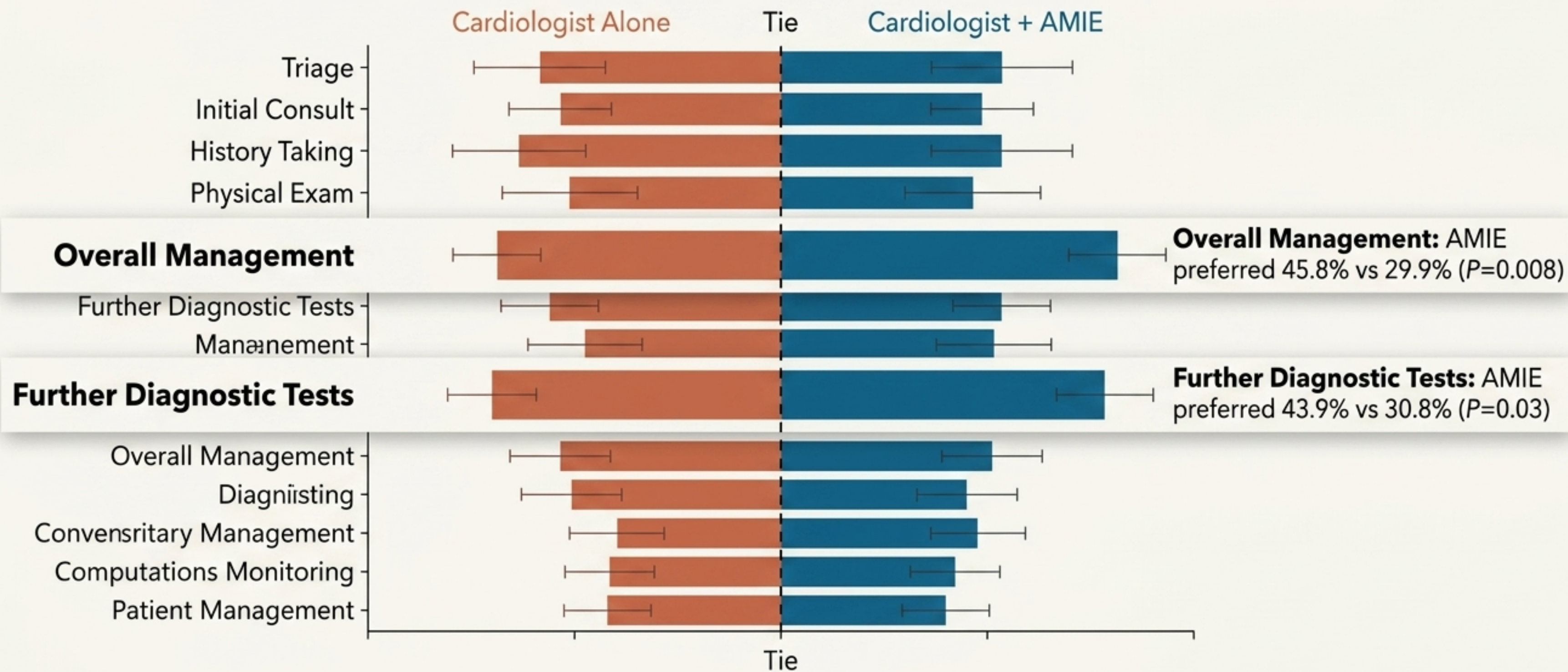
Primary Clinical Outcomes Dashboard

Blinded subspecialists evaluated the paired assessments across ten clinical domains. The augmented approach yielded statistically significant improvements in safety, comprehensiveness, and clinical preference.



Subspecialist Preference by Clinical Domain

While triage and initial consult questions often resulted in a tie, AI-assistance drove significant superiority in the most complex phases of care: determining further diagnostic testing and formulating comprehensive management plans.



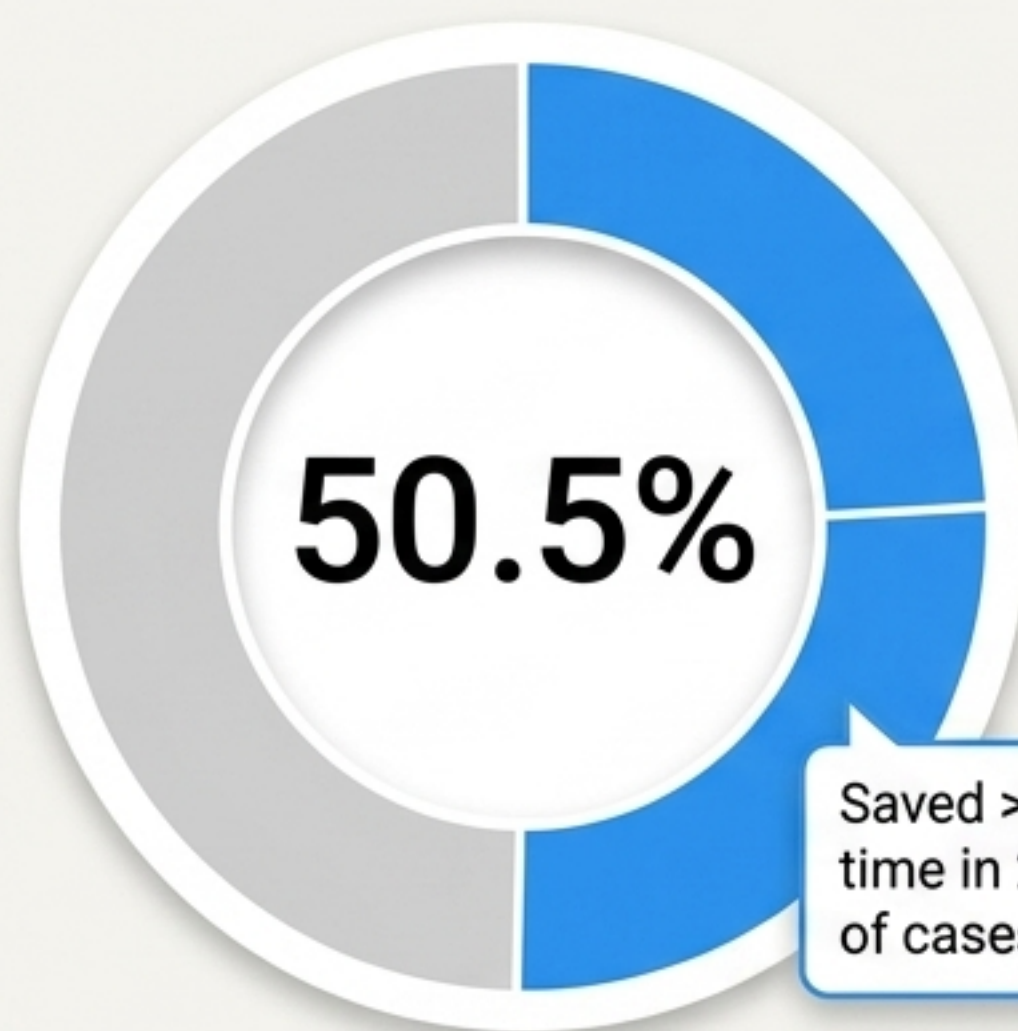
The Augmented Capability Matrix

Subspecialist feedback revealed distinct patterns in how AI assistance transformed the general cardiologists' outputs.

Usual Care (Cardiologist Alone)	AMIE-Assisted Care
Clear and concise clinical reasoning.	Highly comprehensive responses.
Clinically significant omissions in patient history.	Incorporation of modern diagnostics and advanced treatments.
Errors related to key management decisions.	Provoked critical thinking and reconsideration of initial assessments.
Prone to missing complex diagnostic nuances.	Occasional diagnostic overreach or excessive verbosity.

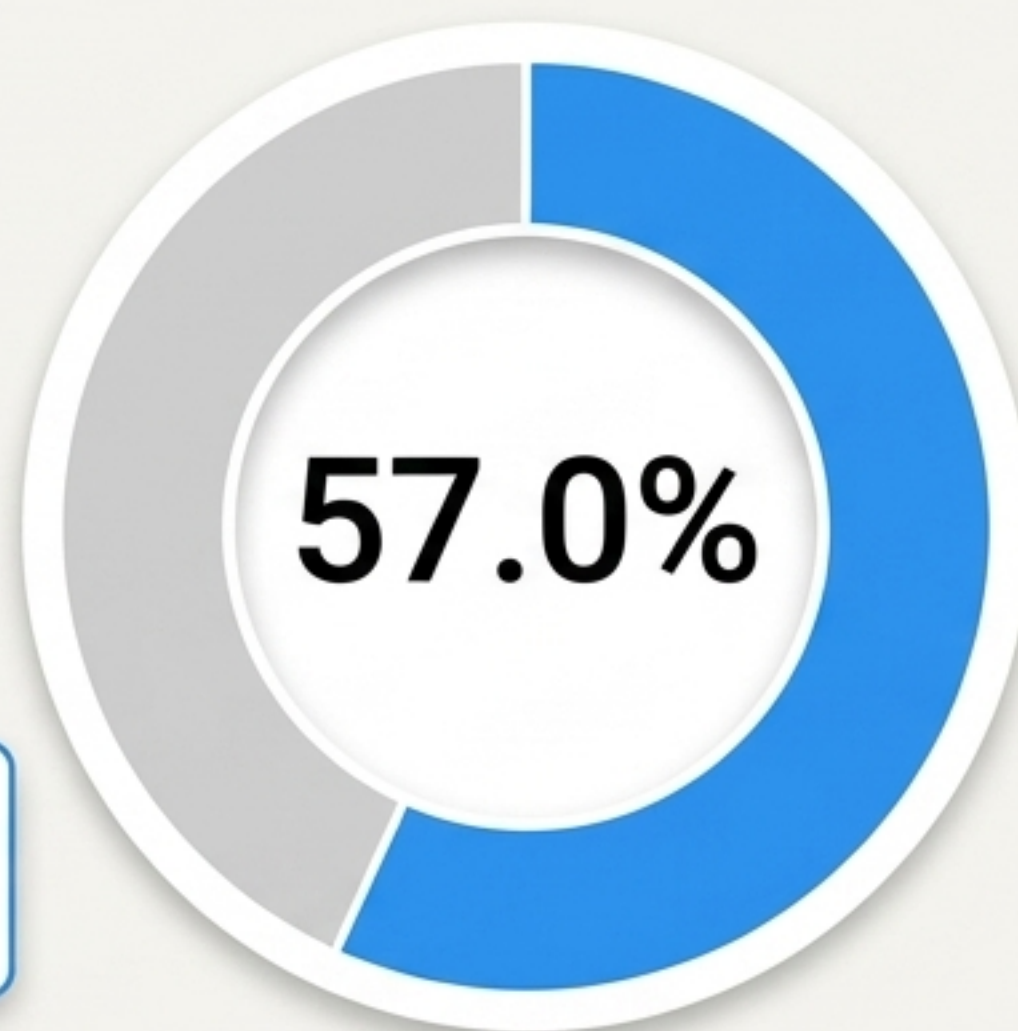
Physician Efficiency and Clinical Confidence

Integrating an LLM into the workflow improved the subjective experience of the participating general cardiologists, delivering time savings alongside cognitive support.

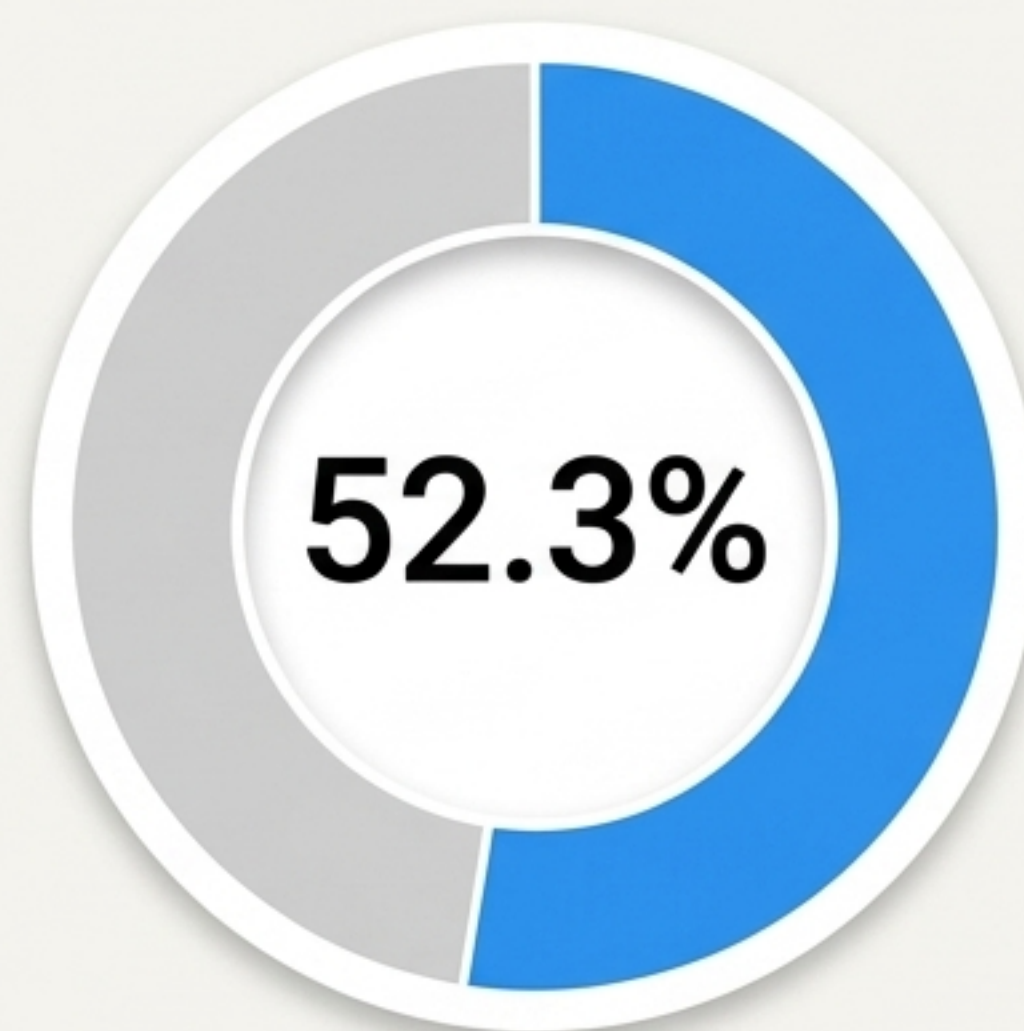


Saved Time

Saved >50%
time in 23.4%
of cases



Helped the Assessment



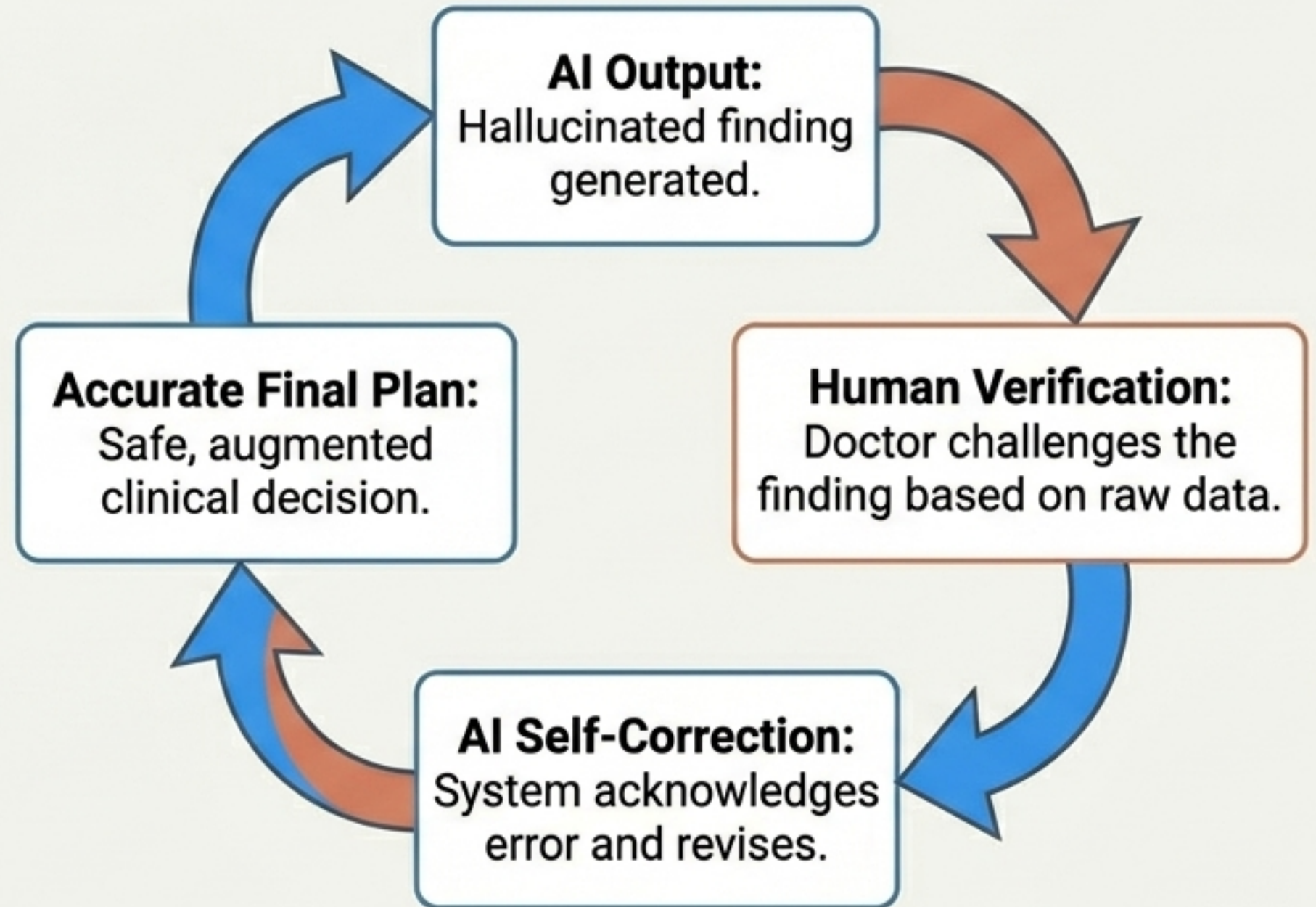
Increased Confidence

Safety Profile: The Hallucination & Correction Loop

Because AMIE is deployed as an interactive assistant rather than an autonomous agent, errors were highly amenable to immediate correction by the physician.

6.5%

Rate of clinically significant hallucinations reported by cardiologists (e.g., fabricating 'left ventricular hypertrabeculation' not present in the source report).



Scaling Subspecialist Expertise

This trial establishes gold-standard, RCT-level evidence that large language models can safely assist generalists in managing complex, life-threatening genetic cardiovascular diseases.

By reducing errors, capturing missing clinical context, and saving time, human-in-the-loop AI systems offer a viable pathway to bridge the global deficit in specialized medical expertise.

